Arid Agriculture University- Gujrat institute of management and sciences

# Computer Organization and Assembly Language

Lab-Assignment #02

**X86 Programming**

**Course Code:**

CSC-211

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**Department:**

Computer Science

**Section:**

CS(A) - General 4th

## Task:

Q1. Write an assembly program to perform addition, subtraction, and multiplication of two numbers using registers and store the results in separate registers.

Q2. Write an assembly program to convert an ASCII digit (e.g., '5') into its corresponding integer value and store the result in a register.

Q3. Write a X86 program to transfer a block of 5 numbers from one memory location to another using loop instructions.

Q4. Write an assembly program to find the sum of 10 numbers stored in an array.

Q5. Write a X86 program to find the largest number in an array of 8 elements.

Q6. Write an assembly program to check whether a number is even or odd using bitwise operations.

Q7. Write a x86 program to print numbers from 1 to 10 using a loop.

Q8. Write an assembly program to multiply two numbers using repeated addition (without using MUL instruction).

Q9. Write a x86 program to calculate the length of a string stored in memory.
Q10.Write a x86 program to swap two numbers Using a temporary registers.

### Q1. Addition, Subtraction, and Multiplication

```
.386

.model flat, stdcall

.stack 4096


.data

   num1 DWORD 10

   num2 DWORD 5


.code

main PROC
```

```
    mov eax, num1      ; eax = 10

    mov ebx, num2      ; ebx = 5


    ; Addition

    mov ecx, eax

    add ecx, ebx       ; ecx = 10 + 5 = 15


    ; Subtraction

    mov edx, eax

    sub edx, ebx       ; edx = 10 - 5 = 5


    ; Multiplication

    mov esi, eax

    imul esi, ebx      ; esi = 10 * 5 = 50


    ret

main ENDP

END main
```

## Output:

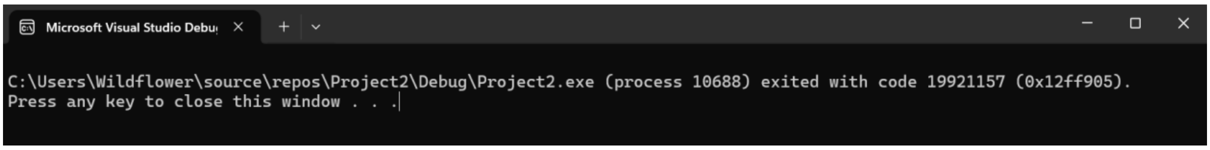
```
C:\Users\Wildflower\source\repos\Project2\Debug\Project2.exe (process 19080) exited with code 10 (0xa).
Press any key to close this window . . .
```

## Q2. ASCII Digit to Integer

```
.386

.model flat, stdcall

.stack 4096
```

```
.data
   asciiChar BYTE '5'  ; ASCII value is 53

.code
main PROC
   mov al, asciiChar   ; al = 53 (ASCII of '5')
   sub al, 30h        ; subtract 48 → al = 5 (actual integer)
   ; Result: al = 5
   ret
main ENDP
END main
```

## Output:




## Q3. Block Transfer Using Loop

```
.386
.model flat, stdcall
.stack 4096

.data
   source DWORD 1, 2, 3, 4, 5
   dest   DWORD 5 DUP(0)      ; 5 empty slots

.code
```

```
main PROC
    mov esi, OFFSET source  ; esi points to source
    mov edi, OFFSET dest    ; edi points to destination
    mov ecx, 5              ; loop counter = 5

transfer_loop:
    mov eax, [esi]          ; load value from source
    mov [edi], eax          ; store value to destination
    add esi, 4              ; move to next DWORD (4 bytes)
    add edi, 4
    loop transfer_loop      ; ecx-- and repeat if ecx != 0

    ret
main ENDP
END main
```

## Output:


## Q4. Sum of 10 Numbers in an Array

```
.386
.model flat, stdcall
.stack 4096

.data
```

```
    arr DWORD 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

    sum DWORD 0



.code

main PROC

    mov esi, OFFSET arr     ; esi = pointer to array

    mov ecx, 10           ; loop 10 times

    mov eax, 0            ; eax = accumulator (sum)



sum_loop:

    add eax, [esi]        ; add current element to eax

    add esi, 4            ; next element

    loop sum_loop



    mov sum, eax          ; store result

    ret

main ENDP

END main
```


## Q5. Largest Number in Array of 8 Elements

```
.386

.model flat, stdcall
```

```
.stack 4096


.data

   arr    DWORD 3, 7, 1, 9, 4, 6, 2, 8

   largest DWORD 0


.code

main PROC

   mov esi, OFFSET arr

   mov ecx, 7            ; compare 7 more times after loading first

   mov eax, [esi]        ; assume first element is largest

   add esi, 4


find_loop:

   mov ebx, [esi]        ; load next element

   cmp eax, ebx          ; compare current largest with next

   jge skip              ; if eax >= ebx, skip update

   mov eax, ebx          ; else update largest


skip:

   add esi, 4

   loop find_loop


   mov largest, eax      ; store largest

   ret
```

```
main ENDP

END main
```

## Output:

```
Microsoft Visual Studio Debu

C:\Users\Wildflower\source\repos\Project2\Debug\Project2.exe (process 15160) exited with code 9 (0x9).
Press any key to close this window . . .
```

## Q6. Even or Odd Using Bitwise AND

```
.386

.model flat, stdcall

.stack 4096


.data

   num    DWORD 7

   result BYTE  "Even", 0   ; default message


.code

main PROC

   mov eax, num

   and eax, 1         ; AND with 1 → checks last bit

                 ; if last bit = 0 → even

                 ; if last bit = 1 → odd

   jz is_even        ; jump if zero flag set (eax AND 1 = 0)
```

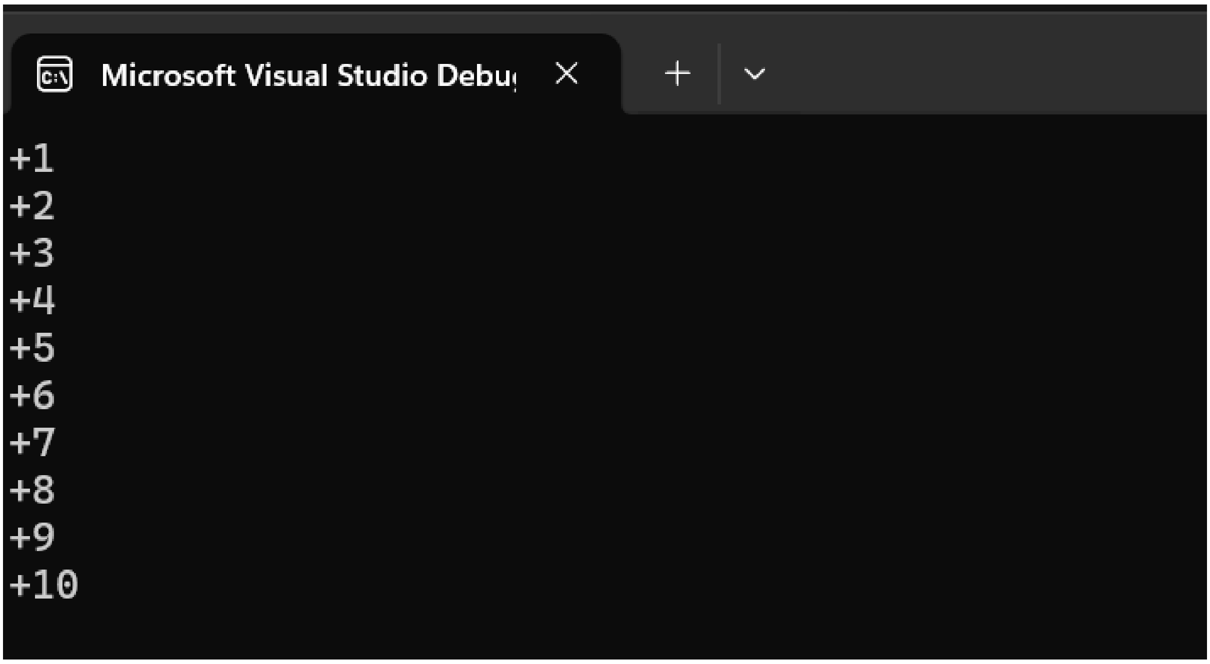
```
   ; Odd case

   ; (you can set a flag or print message here)

   jmp done


is_even:

   ; Even case


done:

   ret

main ENDP

END main
```

## Output:


## Q7. Print Numbers 1 to 10 Using Loop

```
.386

.model flat, stdcall

.stack 4096

ExitProcess PROTO, dwExitCode:DWORD


INCLUDE Irvine32.inc


.code

main PROC

   mov ecx, 10       ; loop 10 times
```

```
    mov eax, 1        ; start from 1

print_loop:
    call WriteInt      ; Irvine32 function: prints integer in eax
    call Crlf          ; new line
    inc eax            ; eax++
    loop print_loop

    INVOKE ExitProcess, 0
main ENDP
END main
```




### Q8. Multiply Using Repeated Addition (No MUL)

```
.386
.model flat, stdcall
```

```
.stack 4096


.data
   num1   DWORD 6
   num2   DWORD 4
   result DWORD 0


.code
main PROC
   mov eax, 0        ; result = 0
   mov ecx, num2      ; loop num2 times (add num1, four times)
   mov ebx, num1


add_loop:
   add eax, ebx       ; eax = eax + num1
   loop add_loop      ; repeat num2 times


   mov result, eax    ; result = 6+6+6+6 = 24
   ret
main ENDP
END main
```

```
Microsoft Visual Studio Debu

C:\Users\Wildflower\source\repos\Project2\Debug\Project2.exe (process 9928) exited with code 24 (0x18).
Press any key to close this window . . .
```

## Q9. Calculate String Length

```
.386
```

```
.model flat, stdcall

.stack 4096


.data

   myStr  BYTE "Hello", 0      ; null-terminated string

   strLen DWORD 0


.code

main PROC

   mov esi, OFFSET myStr       ; esi points to start of string

   mov ecx, 0                  ; counter = 0


count_loop:

   mov al, [esi]               ; load current character

   cmp al, 0                   ; is it null terminator?

   je  done                    ; yes? stop

   inc ecx                     ; no? count it

   inc esi                     ; move to next character

   jmp count_loop


done:

   mov strLen, ecx             ; store length

   ret

main ENDP

END main
```

```
C:\Users\Wildflower\source\repos\Project2\Debug\Project2.exe (process 21252) exited with code 18087168 (0x113fd00).
Press any key to close this window . . .
```

## Q10. Swap Two Numbers Using Temp Register

```
.386
.model flat, stdcall
.stack 4096

.data
    num1 DWORD 25
    num2 DWORD 40

.code
main PROC
    mov eax, num1      ; eax = 25
    mov ebx, num2      ; ebx = 40

    ; Swap using ecx as temp
    mov ecx, eax       ; temp = eax (25)
    mov eax, ebx       ; eax = ebx (40)
    mov ebx, ecx       ; ebx = temp (25)

    ; Now eax = 40, ebx = 25 (swapped!)
    mov num1, eax
    mov num2, ebx

    ret
main ENDP
```

END main

Microsoft Visual Studio Debu

```
C:\Users\Wildflower\source\repos\Project2\Debug\Project2.exe (process 19672) exited with code 40 (0x28).
Press any key to close this window . . .
```